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The tricyclic alkaloid catalyzed crystallization of α-$FAPbI_3$ for high performance antisolvent-free perovskite solar cells†

Zhengzhe Wu,‡[a] Haoyu Cai,‡[a] Tong Wu,[a] Jiayi Xu,[a] Zhenyue Wang,[a] Hongqiang Du,[a] Juan Zhao,[b] Fuzhi Huang, [a] Yi-Bing Cheng [ac] and Jie Zhong *[a]

A high-quality α-phase formamidinium lead triiodide (α-$FAPbI_3$) absorber film is crucial for obtaining perovskite solar cells (PSCs) with outstanding power conversion efficiency (PCE) and stability. An antisolvent-free process exhibits attractive merits for PSC upscaling, while suffering limited solvent evaporation rate, causing substantial difficulty in the precise control of α-$FAPbI_3$ crystallization. In this work, we present a strategy of catalyzed crystallization (CC) of α-$FAPbI_3$ to achieve high performance antisolvent-free PSCs using a tricyclic alkaloid additive, colchicine (Ch). Molecular interactions and *in situ* analysis suggest that the varied functional groups on the tricyclic rings of Ch exhibit specific incorporations with the solvate perovskite precursor, which catalyzed direct formation of α-$FAPbI_3$ compared to the intermediate phase-transition with δ-$FAPbI_3$ and residues occurring in the common routes. Moreover, the Ch molecule *in situ* passivates the grain boundaries and dramatically reduces defects during CC. As a result, the PCE of the CC PSCs reaches 25.06% and the unencapsulated device maintains 83.6% of the initial PCE for more than 1500 h at 85 °C, and 87.6% after 450 h of maximum power point tracking.

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Broader context

Formamidinium lead triiodide ($FAPbI_3$), due to its excellent thermal stability and relatively narrow bandgap of around 1.48 eV, has become one of the most promising absorbers for high performance perovskite solar cells (PSCs). The antisolvent-free process exhibits attractive merits for PSCs upscaling, but it brings about the enormous difficulty of achieving precise phase control. The coexistence of the thermodynamically stable yellow phase δ-$FAPbI_3$ normally requires phase transformation to the black phase α-$FAPbI_3$, which could result in defects and residues at the surface and grain boundaries of the perovskite film, leading to the degrading of the PSCs. Herein, we use a green tricyclic alkaloid molecule, colchicine (Ch), to catalyze the direct formation of α-$FAPbI_3$ from the solvated precursor, without the secondary phase transition in the generation of δ-$FAPbI_3$ and other intermediate phases. As a result, a champion power conversion efficiency (PCE) of 25.06% was obtained. The unencapsulated device demonstrated enhanced stability at 85 °C heating and maximum power point tracking (MPPT) with continuous AM1.5G illumination. Our study provides a convenient strategy to directly obtain a high-quality α-$FAPbI_3$ film with a one-step antisolvent-free process.

1. Introduction

Organo–inorganic metal halide perovskite solar cells (PSCs) have rapidly increased in power conversion efficiency from 3.8% to the present 26.1%, exhibiting great potential for photovoltaic applications.[1,2] Among various perovskite materials, formamidinium lead triiodide ($FAPbI_3$) perovskite has become one of the most promising choices due to its excellent thermal stability and relatively narrow bandgap (E_g = 1.48 eV).[3,4] However, the coexistence of the thermodynamically stable yellow phase δ-$FAPbI_3$ and the black phase α-$FAPbI_3$ during solution deposition disrupts the nucleation and crystallization

[a] *State Key Laboratory of Advanced Technology for Materials Synthesis and Processing, Wuhan University of Technology, Wuhan 430070, P. R. China. E-mail: jie.zhong@whut.edu.cn*

[b] *School of Automobile Engineering, Wuhan University of Technology, Wuhan 430070, P. R. China*

[c] *Foshan Xianhu Laboratory of the Advanced Energy Science and Technology Guangdong Laboratory, Foshan 528216, Guangdong Province, P. R. China*

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‡ Zhengzhe Wu and Haoyu Cai contributed equally.

of the crucial photoactive α-$FAPbI_3$ perovskite.[5] This could result in a large number of defects in the crystals and the grain boundaries of the perovskite film, leading to the degradation of the PSCs. In addition, the relatively over-size of FA^+ induces instability of the $[PbI_6]^{4-}$ framework, which promotes the α-$FAPbI_3$ transforming to the δ-$FAPbI_3$ phase at room temperature.[6] Therefore, achieving controlled crystallization of stable α-$FAPbI_3$ perovskite films remains a great challenge for further improving the power conversion efficiency (PCE) and long-term stability of PSCs.

The most commonly used strategy to control the crystallization with a stabilized α-$FAPbI_3$ lattice is to partially replace FA^+ or I^- with smaller ionic radii ions, such as MA^+, Cs^+, Rb^+, Br^- *etc.*[7–9] However, this tuning of the components of the perovskite leads to band gap broadening and possible localized phase separation.[10] Seok *et al.* found that the addition of volatile alkylammonium chlorides (RACl) could alter the rate of phase transition from δ- to α-$FAPbI_3$, and achieve better crystallinity and superior crystal orientation of the perovskite films.[11] Ionic liquids are also effective in inhibiting the formation of δ-$FAPbI_3$, an intermediate transformation in the crystallization process.[12] In addition, 2D or 3D perovskite seeds to the perovskite precursor solution serve as nucleation sites for enhanced crystallization.[13,14] Alternatively, the use of multifunctional small molecules and polymers by the formation of adducts of additives with $[PbI_6]^{4-}$ inorganic octahedra and the formation of hydrogen bonds with FAI can also modulate the crystallization process and stabilize the α-$FAPbI_3$ perovskite.[15–18]

It is worth noting that dropwise antisolvent processes are normally applied in the α-$FAPbI_3$ perovskite preparation. The film quality will be limited by the precise time window of the antisolvent dripping,[19] and it is an enormous challenge to prepare uniform large-area perovskite films, which inhibits further upscaling of PSCs.[20] Moreover, the residue impurity phase resulting from the complex intermediate phase of the commonly used dimethylsulfoxide (DMSO) could lead to potential instability of the devices.[21,22] Antisolvent-free solvents are preferred for perovskite deposition while requiring extra efficient and delicate crystallization control. Solvent mixtures, such as 2-ME and CHP,[23] DMA and RNH_3Cl ethanol,[24] and DMF and NMP,[25] incorporation with an ionic liquid,[26–28] and adjustment of the perovskite constitution[29] are reported to meditate intermediate phase and α-$FAPbI_3$ crystallization. The highest PCE for antisolvent-free $FAPbI_3$ PSCs reaches 25.1%.[24] Therefore, it is of great importance to further develop an efficient and convenient route to control the α-$FAPbI_3$ crystallization with an antisolvent-free process.

In this work, we report a novel catalyzed crystallization (CC) to fabricate a high-quality pure-phase α-$FAPbI_3$ perovskite films with a tricyclic alkaloid, colchicine (Ch). The α-$FAPbI_3$ is directly formed in antisolvent-free spin-coating without intermediate phase transformation under the catalysis reaction of Ch molecules. In contrast, the sample without CC demonstrates obvious δ- to α-phase transformation and precipitation of PbI_2. We found that the various functional groups on the Ch rings benefit the interaction with the solvated perovskite species and play the role of catalysts for enhanced nucleation and proper crystal growth of α-$FAPbI_3$. This induces *in situ* defect passivation with Ch, which significantly improves the device PCE and stability. As a result, a champion PCE of the device fabricated through the CC process showed a significant enhancement of the PCE to 25.06% with a V_{OC} of 1.176 V, and fill factor (FF) of 83.6%, compared to the PCE of 23.49% for the control sample. The unencapsulated devices retained about 83.6% of the initial PCE after 1500 h of thermal aging at 85 °C under N_2. The encapsulated device retained 87.6% of its initial PCE after 450 h of continuous illumination at the maximum power point under atmospheric conditions.

2. Results and discussion

Colchicine, derived from the bulb of the *Colchicum autumnale L.*, is a very useful alkaloid for mutagenic plant polyploid, anti-inflammatory, anti-tumor, and gout treatment. We present a catalyzed crystallization (CC) with it (CC–Ch) to fabricate high-quality α-$FAPbI_3$ perovskite films. Fig. 1(a) shows the tricyclic molecular structure of Ch, consisting of a carbonyl group (C═O), an amide group (–CONH–), and four ether bonds (C–O–C). The calculated electrostatic potential (ESP) of Ch (Fig. 1(a)) shows the negative charges located on the amide oxygens and the carbonyl oxygens on the seven-membered ring (blue region), while the amide hydrogen is electropositive (red region), which implies potential interaction with Pb^{2+}/FA^+ and I^-, respectively. The calculated interaction energy between Ch and PbI_2 is −96.95 kcal mol^{-1}, −292.97 kcal mol^{-1} for Ch/FA^+, and −305.39 kcal mol^{-1} for Ch/I^- (Fig. 1(a)). Further analysis of the calculations revealed the existence of CONH···Pb coordination bonds, N–H···I hydrogen bonding interaction forces between Ch and PbI_2, and strong hydrogen bonding interactions between Ch (C═O) and FAI (N–H) (Fig. S1, ESI†).

We further confirmed the effective interactions between Ch and solvated perovskite species through experiments. UV-vis absorption measurements were carried out for solutions of FAI, PbI_2, and perovskite with varied Ch contents (Fig. 1(b)–(d)). The pure FAI solution showed a sharp and symmetric I^- signal peak at 260 nm wavelength, and two broad peaks at 295 and 366 nm are assigned for I_2 and I_3^-, respectively (Fig. 1(b)). With the increase of the Ch content, the peak of I^- gradually red-shifted, while the peaks of I_2 and I_3^- disappeared. This suggests that the interaction between Ch and FAI and the aromatic ring on Ch can capture trace amounts of I_2.[30] For the PbI_2 solution, the absorption peaks of PbI^+, PbI_2, and PbI_3^- appeared at 278, 323, and 368 nm,[31,32] respectively (Fig. 1(c)). The absorption peaks of PbI^+, PbI_2, and PbI_3^- appeared at 278, 324, and 370 nm, respectively, in the solution of the perovskite precursor without Ch additive (Fig. 1(d)). When 5% Ch was added to the PbI_2 solution, the PbI_3^- peak weakened and the PbI_2 peak was slightly enhanced. Similar results were observed in the precursor solution. The PbI^+ peak was slightly blue-shifted with the disappearance of PbI_3^- and the significant enhancement of the PbI_2 peak. This indicates that Ch could form complex adducts

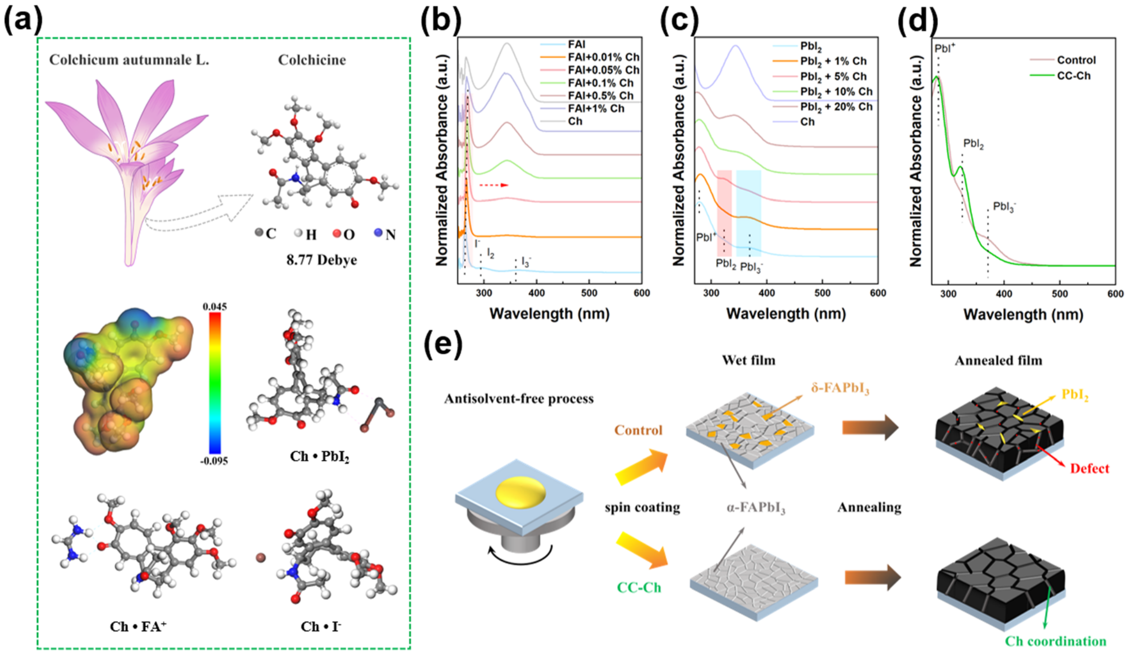

Fig. 1 (a) Molecular structure, electrostatic potential, and the interactions between Ch and $PbI_2/FA^+/I^-$ of colchicine (Ch). UV-vis absorption spectra of (b) FAI/Ch solutions, (c) PbI_2/Ch solutions, and (d) perovskite precursor solutions with different molar ratios of Ch additive. (e) Schematic diagram of the catalyzed crystallization with Ch for one-step antisolvent-free preparation of α-$FAPbI_3$ perovskite films.

with PbI_2 and FAI in the solution, and reduce the interaction between the iodine ion and lead. Therefore, these multiple interactions promise Ch a crucial role in the crystallization of α-$FAPbI_3$ with the one-step antisolvent-free process. The CC–Ch process forms large grains, exhibits *in situ* defect passivation, and is free of PbI_2 particles on the surface, as illustrated in Fig. 1(e).

To precisely determine the functions of the groups on the colchicine molecule, we investigated the interactions between Ch and the perovskite by liquid nuclear magnetic resonance (NMR) spectroscopy, X-ray photoelectron spectroscopy (XPS), and Fourier transform infrared (FTIR) spectroscopy, as shown in Fig. 2. Fig. 2(a) and (b) show the full and localized magnification ^{1}H NMR spectra of Ch, FAI, and their mixtures in deuterated DMSO solvent, respectively. The results showed that the resonance signals of the protonated ammonium in FAI were two broad peaks (8.69 and 8.95 ppm), and the resonance signal on Ch showed one sharp peak (7.86 ppm). After the addition of Ch, the broad peak at 8.95 ppm of FAI becomes sharp and shifts to 9.00 ppm in the downfield direction. The peak at 8.69 ppm splits into two and the one at 7.86 ppm splits into multiple peaks. These results indicate that the chemical environment of the active hydrogen in FA^+ has been changed and a strong hydrogen bond between Ch and FA^+ was formed.[18,33] The characteristic peaks of Ch at 8.56 and 8.58 ppm, which can be attributed to N–H, slightly shifted to 8.55 and 8.57 ppm, which could be attributed to the redistribution of electron cloud density due to the hydrogen bonding between Ch and I (Fig. S1, ESI†). This conclusion is further supported by the changes in the ^{1}H NMR spectra of the Ch and Ch + PbI_2 mixtures, as shown in Fig. S2 (ESI†) and Fig. 2(c). To verify the interaction between Ch and PbI_2, the ^{13}C NMR spectra of Ch and Ch + PbI_2 mixtures in deuterated dimethyl-sulfoxide are compared in Fig. 2(d). The results showed that the peak of Ch at CONH (position 1) shifted from 168.52 to 168.51 ppm. However, the ^{13}C NMR spectra of Ch, FAI, and their mixtures (Fig. 2(e)) show that the chemical shift of the amide group C═O bond (position 1) on Ch remains unchanged and the chemical shift of the C═O bond (position 2) on the seven-membered carbocyclic ring shifts from 178.00 to 177.99 ppm. These distinct results suggest that interactions existed between Pb^{2+} and the amide oxygen (position 1) and the interaction between C═O (position 2) and FA^+. The chemical shift of –CH (position 3) on FAI shifted in the downfield direction (157.15 → 157.19 ppm), reducing the shielding effect, which again proved the interaction with FAI and C═O (position 2).[34] According to the calculation (Fig. S1, ESI†) these two interactions can be ascribed to the coordination bond and hydrogen bond, respectively, for C═O at position 1 and position 2 on the Ch molecule.[35]

XPS analysis is conducted to further analyze the detailed interactions and the full spectra are shown in Fig. S3 (ESI†). The binding energy peaks of Pb $4f_{5/2}$ and Pb $4f_{7/2}$ are located at 143.23 and 138.35 eV, respectively, as shown in Fig. 2(f). With the addition of Ch, these peaks move towards the direction of lower binding energies to 142.92 and 138.06 eV, and the small peaks of metal Pb^0 close to the main peaks have almost disappeared, which implies an enhancement of the density of the electron cloud around lead.[36] The shift of the Pb 4f peak suggests the existence of a coordination interaction between C═O and Pb^{2+} and prevents the formation of metal Pb^0, which

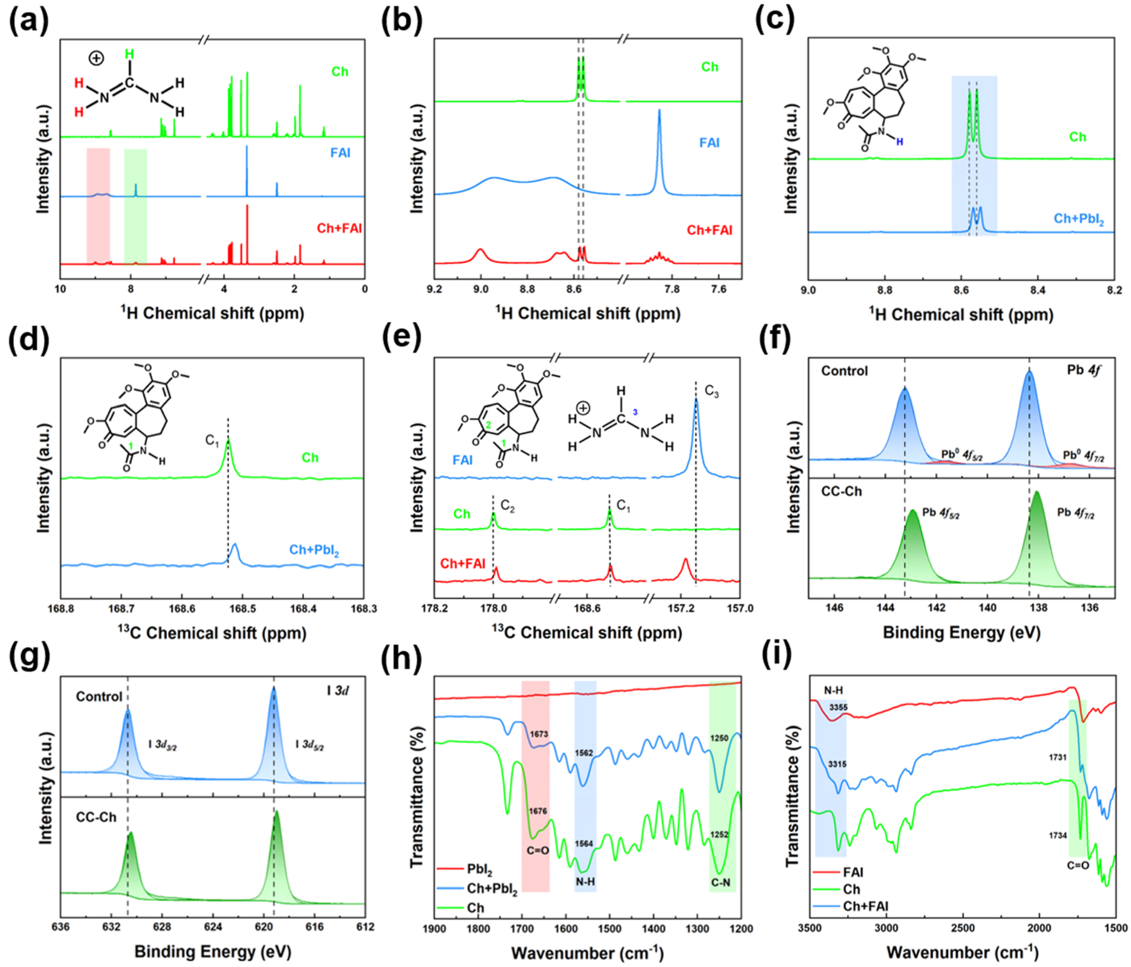

Fig. 2 (a) ^{1}H NMR spectra of Ch, FAI, and their mixed solutions. (b) ^{1}H NMR localized magnification spectra from the highlighted areas of (a). (c) ^{1}H NMR spectra of Ch and Ch + PbI_2 solutions. (d) ^{13}C NMR spectra of Ch and Ch + PbI_2 solutions. (e) ^{13}C NMR spectra of Ch, FAI, and their mixed solutions. XPS spectra of (f) Pb 4f, and (g) I 3d of perovskite films with or without Ch additive. FTIR spectra of (h) powder of Ch, PbI_2, Ch + PbI_2 and (i) powder of Ch, FAI, Ch + FAI.

normally retards the decomposition process of the perovskite. A similar trend is observed for the binding energy of I (Fig. 2(g)). After Ch modification, the peaks corresponding to I $3d_{3/2}$ and I $3d_{5/2}$ are shifted from 630.65 and 619.21 eV in the pristine perovskite films to 630.42 and 618.98 eV, respectively. This indicates an interaction between FAI and Ch, which is consistent with the NMR results.

The interactions between Ch and perovskite were further investigated by FTIR, as shown in Fig. 2(h). Upon the addition of PbI_2 powder, the C=O stretching vibrational peak on Ch was shifted from 1676 to 1673 cm^{-1}, indicating the creation of a CONH···Pb coordination bond between Ch and PbI_2. This also further validates the ^{13}C NMR conclusion about the formation of coordination bonds between Pb^{2+} and amide oxygen. Similarly, the N–H bending vibrational peaks (1564 cm^{-1}) and C–N stretching vibrational peaks (1252 cm^{-1}) on Ch were shifted to lower wavenumbers of 1562 and 1250 cm^{-1}, respectively, and the N–H bending vibrational peaks changed from broad to sharp peaks, which can be attributed to the generation of interactions between Ch and I.[37] This further confirms NMR observations and calculations that a hydrogen bond is formed between I and N–H on the amide group of Ch. When FAI and Ch are mixed, the N–H stretching vibrational peak (3355 cm^{-1}) on FAI becomes narrower and stronger shifted to 3315 cm^{-1} and the C=O stretching vibrational peak (1734 cm^{-1}) of the seven-membered carbon ring on Ch becomes weaker and shifted to 1731 cm^{-1} (Fig. 2(i)). This provides strong evidence for the formation of N–H···C=O hydrogen bonding interactions between FAI and Ch. Thus, we speculate that the addition of Ch additive can play a critical role in regulating the nucleation and crystallization process of the perovskite through the hydrogen bonding interactions between FAI (N–H) and Ch (C=O), the coordination bonding between PbI_2 and Ch (CONH), and the hydrogen bonding interactions between I and Ch (N–H).

To obtain the conceivable mechanism of Ch for perovskite crystallization, we performed *in situ* optical microscopy, *in situ* UV-vis absorption spectroscopy, and *in situ* XRD to monitor the evolution process from the precursor solution to the perovskite films. The optical microscopy results suggest that perovskite

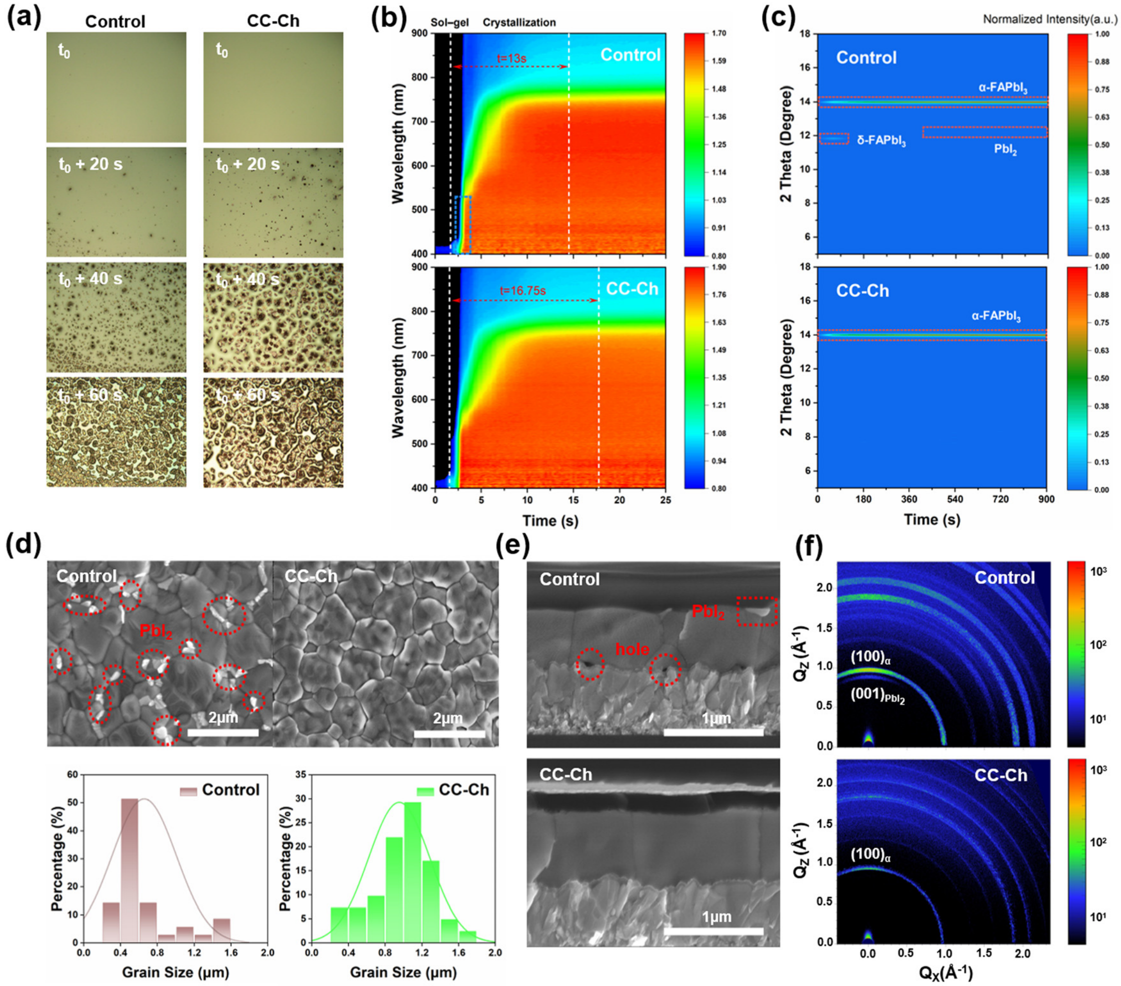

Fig. 3 (a) The optical microscopy of the growth of perovskite crystals from precursor films. (b) *In situ* UV-vis absorption spectra during the initial annealing process. (c) *In situ* XRD spectra of the perovskite precursor solutions during annealing after spin coating. (d) Top-view SEM images and histogram plots of the grain size distributions of the perovskite films. (e) Cross-sectional SEM images and (f) 2D GIWAXS patterns of the perovskite films with and without Ch additive.

precursor solutions with and without Ch all started to form perovskite nuclei from t_0, and the growth was completed within t_0 + 60 s (Fig. 3(a)). It is noteworthy that the addition of Ch allows the precursor to form a large number of uniform hexagonal nuclei in a short period. As shown in Fig. 3b and Fig. S4 (ESI†), the *in situ* UV-vis absorption intensity of the perovskite films increased from annealing for about 2 s, indicating the formation of perovskite crystals. The crystallization process of the perovskite films with Ch additive is 16.75 s, which is longer than that of the control films (13 s), with enhanced absorption intensity compared to the control. This suggests that the Ch additive retarded the growth rate of the perovskite grains, which may be due to the strong interactions between Ch and the perovskite components. It is interesting to note that the perovskite films without Ch have an absorption threshold at 400 nm at around 2.5 s, which is not present in the perovskite films with Ch additives (Fig. S5, ESI†). This could be the absorption peak caused by the precipitation of excess PbI_2.[38] The *in situ* XRD results (Fig. 3(c)) show that, despite faster crystallization, both δ-phase (2θ = 11.8°) and α-phase (2θ = 14.0°) perovskite appeared in the control sample. The δ-phase transformed into the α-phase with continuous annealing after 120 s. However, the diffraction signal peak of PbI_2 appeared and gradually enhanced with the annealing time. In a distinct contrast, the sample with Ch additives demonstrated the direct formation of α-$FAPbI_3$ without the phase transition of the δ-phase and formation of PbI_2.

The scanning electron microscopy (SEM) morphologies, as shown in Fig. 3(d), confirmed that a considerable amount of white PbI_2 particles existed on the surface of the control with an average grain size of ~650 nm. The CC–Ch perovskite films became more homogeneous without the appearance of the PbI_2 particles and the average grain size up to ~950 nm. The grain size of the perovskite becomes smaller when the Ch content is gradually increased (Fig. S6, ESI†). This may be attributed to the strong interaction of Ch within the perovskite grain boundaries, which hinders grain growth. The root-mean-square roughness (RMS) of the surface determined by atomic force microscopy (AFM) was significantly reduced from 68.43 nm to 50.97 nm on a 20 × 20 μm^2 area with the addition of Ch (Fig. S7, ESI†). The contact angle of the perovskite films becomes progressively smaller when the Ch content is gradually increased, which is consistent with the SEM findings (Fig. S8, ESI†). The cross-section SEM image (Fig. 3(e)) suggests that pinholes and white PbI_2 particles were generated in the perovskite films, whereas the CC–Ch perovskite formed compact aligned large grains with a smooth surface, which is more favorable for charge transport at the interface. This observation further implies the critical role that Ch played in the perovskite crystallization. The surface of the perovskite films was investigated by grazing incidence wide angle scattering (GIWAXS) (Fig. 3(f) and Fig. S9, ESI†). GIWAXS with a grazing incidence angle of 0.1° can probe the surface layer of the perovskite films to a depth of about 30 nm. The PbI_2 signal was observed in the perovskite films without Ch (q = 0.888 Å^{-1}), while no PbI_2 signal was shown in the CC–Ch sample. This further validates the conclusions of effective crystallization mediation of Ch in the antisolvent-free perovskite preparation.

The proposed catalyzed crystallization process of Ch for α-$FAPbI_3$ is presented in Fig. 4. Generally, during the crystallization of $FAPbI_3$ perovskite, the α-phase and δ-phase exist simultaneously, and the phase transformation is promoted by the thermal derived ion diffusion in the annealing step. Due to the larger size of FA^+, it is more difficult to enter the laminar structure of PbI_2, and thus FAI and PbI_2 are more likely to form δ-$FAPbI_3$ with a low formation energy barrier.[39–41] The incorporation of Ch alters the interactions between PbI_2 and FAI in the solution of the perovskite precursor, with the formation of strong coordination interactions between Ch (C=O, red) and Pb^{2+}, hydrogen bonding between Ch (C=O, green) and FA^+, and hydrogen bonding forces between Ch (N–H, blue) and I^-. The varied interactions reduce the energy required to convert PbI_2 into the α-$FAPbI_3$ perovskite, thereby assisting the direct nucleation and crystallization of α-$FAPbI_3$ without the transition of δ-$FAPbI_3$.[6,26,38] It needs to be noted that there is no intermediation phase generated with the Ch, making the Ch molecule a catalyst for the direct formation of photoactive α-$FAPbI_3$. Favorably, the residue Ch molecules still existed as the passivation for defects and inhibited the formation of PbI_2 particles, which dramatically improved the quality of the perovskite films.

To investigate the effect of the catalysis crystallization, the energy band structure of the device is illustrated in Fig. 5(a). The absorption intensity was slightly enhanced with CC–Ch and the band gaps of the perovskite maintained the same at ∼1.54 eV (Fig. S10, ESI†) as the control sample. The ultraviolet photoelectron spectroscopy (UPS) tests suggest that the work function (W_F) and valence band maximum (VBM) were calculated based on the secondary electron cutoff band ($E_{cut\text{-}off}$) and the Fermi energy level ($E_{F\text{-}edge}$) of the perovskite films (Fig. S11, ESI†). The conduction band minimum (CBM) was also calculated based on the tac diagram of the UV-vis absorption spectrum. According to the calculated results (Table S1, ESI†), the W_F of the CC–Ch perovskite was elevated by 0.14 eV (from −4.15 to −4.01 eV). The valence bands (E_{VB}) of both the control and the perovskite films with Ch were −5.59 and −5.40 eV, respectively. The energy level gap (ΔE) for the hole transfer between the perovskite and Spiro-OMeTAD was reduced from 0.39 to 0.20 eV after the addition of Ch. This band shifting improves the energy level alignment, which facilitates the extraction and transport of holes from the perovskite to the (hole transport layer) HTL, and increases the V_{OC}

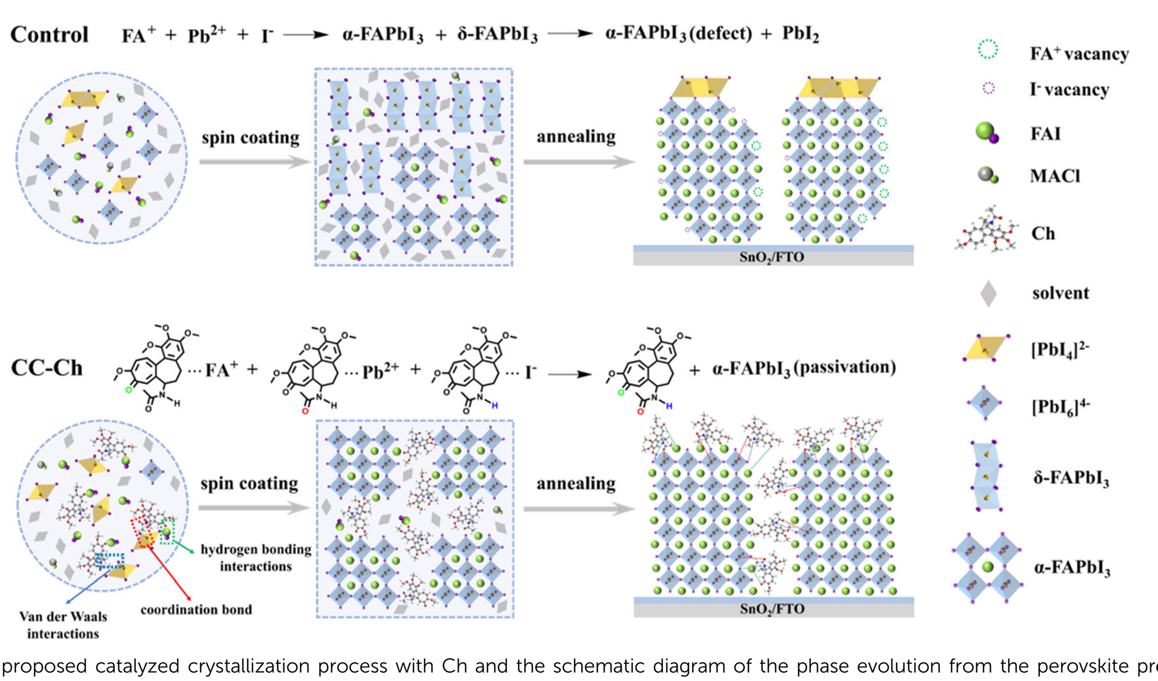

Fig. 4 The proposed catalyzed crystallization process with Ch and the schematic diagram of the phase evolution from the perovskite precursor to α-$FAPbI_3$ perovskite films.

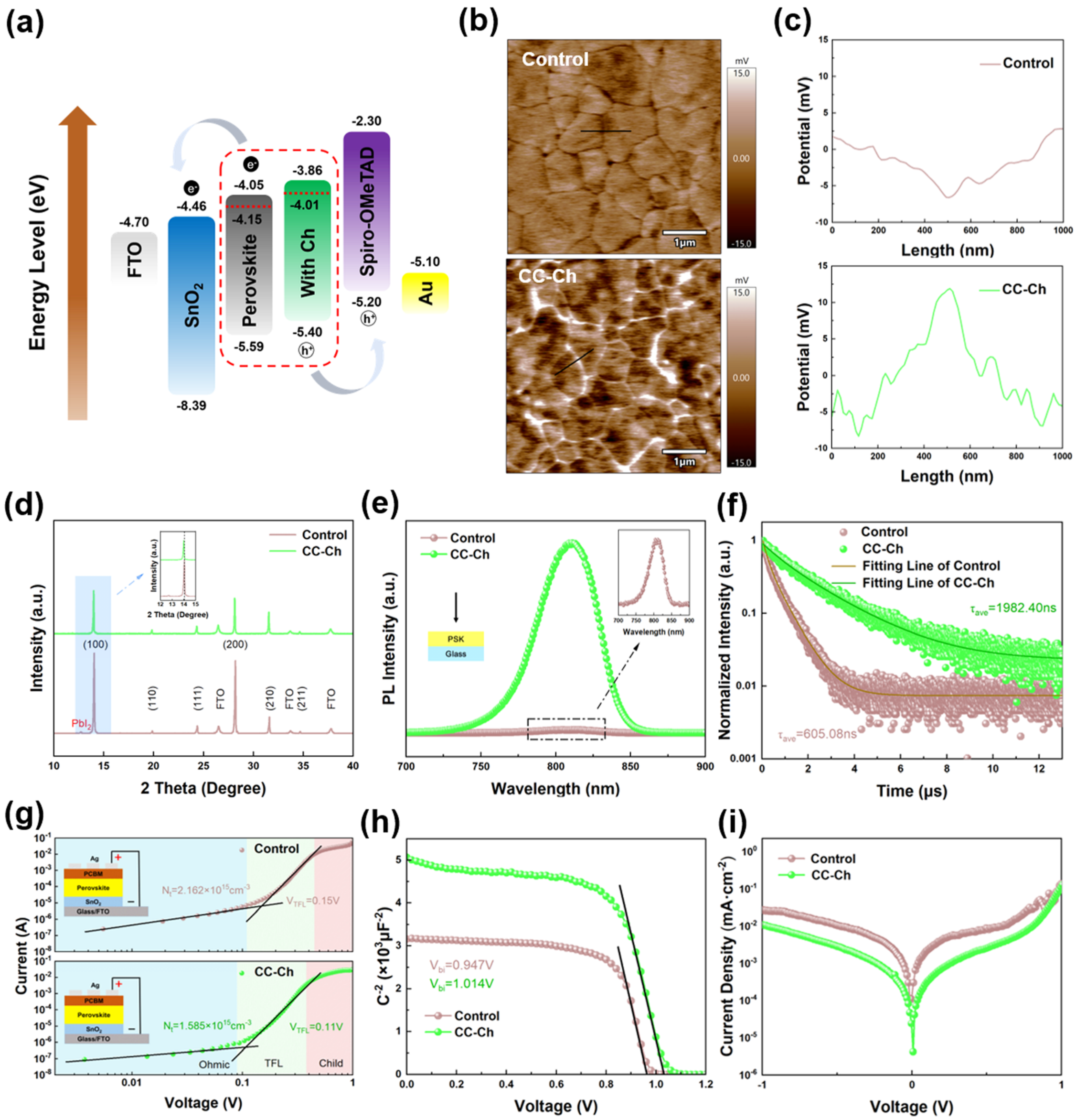

Fig. 5 (a) Energy level diagram of the control and target devices. (b) KPFM surface potential images of the perovskite films with and without Ch additive, and the black line is the path for quantitative measurement of the potential. (c) Line scan images of the potential difference between the grain boundary and grain interior from (b). (d) XRD spectra of the perovskite films with and without Ch additive. (e) Steady-state PL (the inset figure: perovskite films without Ch additive) and (f) TRPL spectra of perovskite films with and without Ch additive deposited on glass. (g) Dark *J*–*V* curves of the electron-only devices: FTO/SnO_2/perovskite/PCBM/Ag. (h) Mott–Schottky curves and (i) dark *J*–*V* curves of the devices with and without Ch.

of the device.[42] The surface potentials of the perovskite films were measured using Kelvin probe force microscopy (KPFM) (Fig. 5(b)). The surface potential was scanned linearly at a randomly taken grain boundary, and the tip of the needle was passed through the grain interiors (GIs) and grain boundaries (GBs) sequentially, as shown in Fig. 5(c). The results show that the potentials of the GBs of the perovskite films of the pristine samples are lower than those of the GIs. On the contrary, the potential of the GBs was significantly higher than that of the GIs for the perovskite films after the addition of Ch. This is probably due to the Ch located at the grain boundaries, which passivates the GBs and increases the contact potential difference at the grain boundaries.[43,44]

The diffraction peaks of PbI_2 disappeared after the addition of Ch and the diffraction intensity of the peaks in the (100) and (200) planes decreased with the increase of the Ch content (Fig. 5(d) and Fig. S12, ESI†). At the same time, they are gradually shifted to a lower angle with the full-width-half-magnitude (FWHM) of the peaks in the (100) plane decreasing from 0.102° to 0.091° (Table S2, ESI†). This is attributed to the fact that Ch can mitigate the strong preferred orientation induced by MACl, which reduces the external stresses in the perovskite films and thus reduces the defect density.[45] Steady-state photoluminescence (PL) and time-resolved photoluminescence (TRPL) spectroscopy were used to investigate the defect related carrier dynamics of the perovskite films. The perovskite

Table 1 The champion and (average) parameters of the PSCs doped with different molar ratios of Ch

Concentration	V_{OC} (V)	J_{SC} (mA cm^{-2})	FF	PCE (%)
Control	1.165 (1.140 ± 0.012)	24.92 (25.12 ± 0.18)	0.809 (0.792 ± 0.011)	23.49 (22.68 ± 0.46)
0.01%	1.138 (1.132 ± 0.005)	25.36 (25.10 ± 0.16)	0.811 (0.807 ± 0.015)	23.39 (22.92 ± 0.53)
0.05%	1.176 (1.175 ± 0.010)	25.49 (25.27 ± 0.11)	0.836 (0.827 ± 0.008)	25.06 (24.56 ± 0.24)
0.1%	1.157 (1.135 ± 0.011)	24.90 (25.00 ± 0.15)	0.800 (0.796 ± 0.005)	23.16 (22.60 ± 0.31)
0.5%	1.042 (1.029 ± 0.012)	24.17 (23.95 ± 0.28)	0.610 (0.557 ± 0.032)	15.44 (13.76 ± 0.98)
1%	1.004 (0.998 ± 0.005)	22.65 (22.95 ± 0.16)	0.435 (0.416 ± 0.011)	9.90 (9.54 ± 0.23)

films with Ch exhibited a significantly enhanced PL intensity compared to the control (Fig. 5(e)), indicating significantly reduced defects due to the catalyzed crystallization kinetics of the Ch perovskite films. By fitting a double exponential function to the TRPL curves, it can be obtained that the average carrier lifetime of the perovskite films with Ch (1982.40 ns) is much higher than that of the control sample (605.08 ns) (Fig. 5(f) and Table S3, ESI†). These results demonstrate that CC–Ch effectively reduces nonradiative recombination by coordination with perovskite films.

We further verified the effectiveness of Ch in defects suppressed by space charge-limited current (SCLC) measurements. The electronic-only device of FTO/SnO_2/perovskite/PCBM/Ag was fabricated to calculate the electron trap density within the perovskite films (Fig. 5(g)). The trap density can be found by the formula: $N_t = (2\varepsilon\varepsilon_0 V_{TFL})/(eL^2)$, where ε, ε_0, e, and L are the relative permittivity of the perovskite (46.9),[46] vacuum permittivity (8.85×10^{-14} F cm^{-1}), electron charge (1.6×10^{-19} C), and thickness of the perovskite films (600 nm). The V_{TFL} of the control and devices with Ch was 0.15 and 0.11 V, respectively, with corresponding calculated trap densities of 2.162×10^{15} and 1.585×10^{15} cm^{-3} (Table S4, ESI†). The reduction of the electronic trap state density indicates that Ch effectively passivates the defects and reduces the nonradiative recombination of the carriers. The built-in potential (V_{bi}) of the devices was analyzed by measuring the Mott–Schottky curve at 10 kHz (Fig. 5(h)). The V_{bi} of the device with Ch is 1.014 V, which is significantly larger than the control 0.947 V. This suggests improved driving force for charge separation, consistent with the trend of V_{OC} enhancement in the J–V curve. The devices with Ch exhibited lower dark current densities and higher slopes compared to the control samples (dark J–V curves in Fig. 5(i)), suggesting less interfacial carrier recombination and current leakage in the devices.

The photovoltaic performance of the devices with CC–Ch additives was evaluated by constructing the planar devices of FTO/SnO_2/perovskite/Spiro-OMeTAD/Au. As shown in Fig. S13 (ESI†), J–V curves were obtained for champion efficiency devices with different molar ratios of Ch (0.01%, 0.05%, 0.1%, 0.5%, and 1%) under standard AM1.5G illumination. The corresponding detailed photovoltaic parameters are summarized in Table 1 and Fig. S14 (ESI†). The control device showed the highest PCE of 23.49%, V_{OC} of 1.165 V, short-circuit current density (J_{SC}) of 24.92 mA cm^{-2}, and FF of 80.9%. The CC–Ch device shows a champion PCE of up to 25.06% (parameters of different devices shown in Table S6, ESI†), V_{OC} of 1.176 V, J_{SC} of 25.49 mA cm^{-2}, and FF of 83.6% (Fig. 6(a)). The significantly improved optoelectronic performance can be attributed to the improved crystallization process of CC–Ch with high-quality perovskite films and low nonradiative defect density. The histogram of PCE statistics (Fig. 6(b), 30 devices each) illustrates an average PCE of 24.35% and 22.87% for devices with and without Ch, respectively, showing the significant reproducibility and reliability of PSCs with CC–Ch.

We depict the external quantum efficiency (EQE) plot and the integrated current density of the EQE curve in Fig. 6(c). The integrated current densities of the control and the device with Ch are 24.92 and 25.04 mA cm^{-2}, respectively, which are in good agreement with the J_{SC} values in the J–V curves. Interestingly, the absorption intensity of the devices with Ch is slightly increased below 400 nm, which indicates that the addition of Ch improves the charge transport and carrier collection in the SnO_2/perovskite interface.[47] The arcs in the high-frequency region reflect the charge-transfer resistance (R_{ct}), and a smaller R_{ct} in the electrochemical impedance spectroscopy curves (EIS) (Fig. 6(d)) indicates a higher electron extraction capability. The R_{ct} of the device decreased from 5714 to 2678 Ohm after the addition of Ch, which reflects faster carrier transport and lower recombination rates in the CC–Ch PSCs (Table S5, ESI†). The measured V_{OC} of the device as a function of the light intensity can be applied to study the kinetics of charge extraction and recombination (Fig. 6(e)). The ideal factor (n) can be obtained by fitting V_{OC} to the logarithmic light intensity ($\ln\Phi$): $V_{OC} = n_{id}k_BT\ln(\Phi)/q + b$, where k_B is the Boltzmann constant, T is the temperature, Φ is the light intensity, and q is the elementary charge, which can be extracted from the slope of the curve with the ideal factor n_{id}. The closer n_{id} is to 1, the smaller the trap-assisted Shockley Read Hall (SRH) composite process is. Compared to the control ($1.624k_BT/q$), the slope of the device with Ch is smaller ($1.369k_BT/q$). This further confirmed passivated deep defects with Ch for higher V_{OC}. Steady-state power output (SPO) was performed to evaluate the operational stability of the PSCs at the maximum power point (MPP) of the J–V test for 300 s (Fig. 6(f)). The output power of the PSCs with Ch hardly

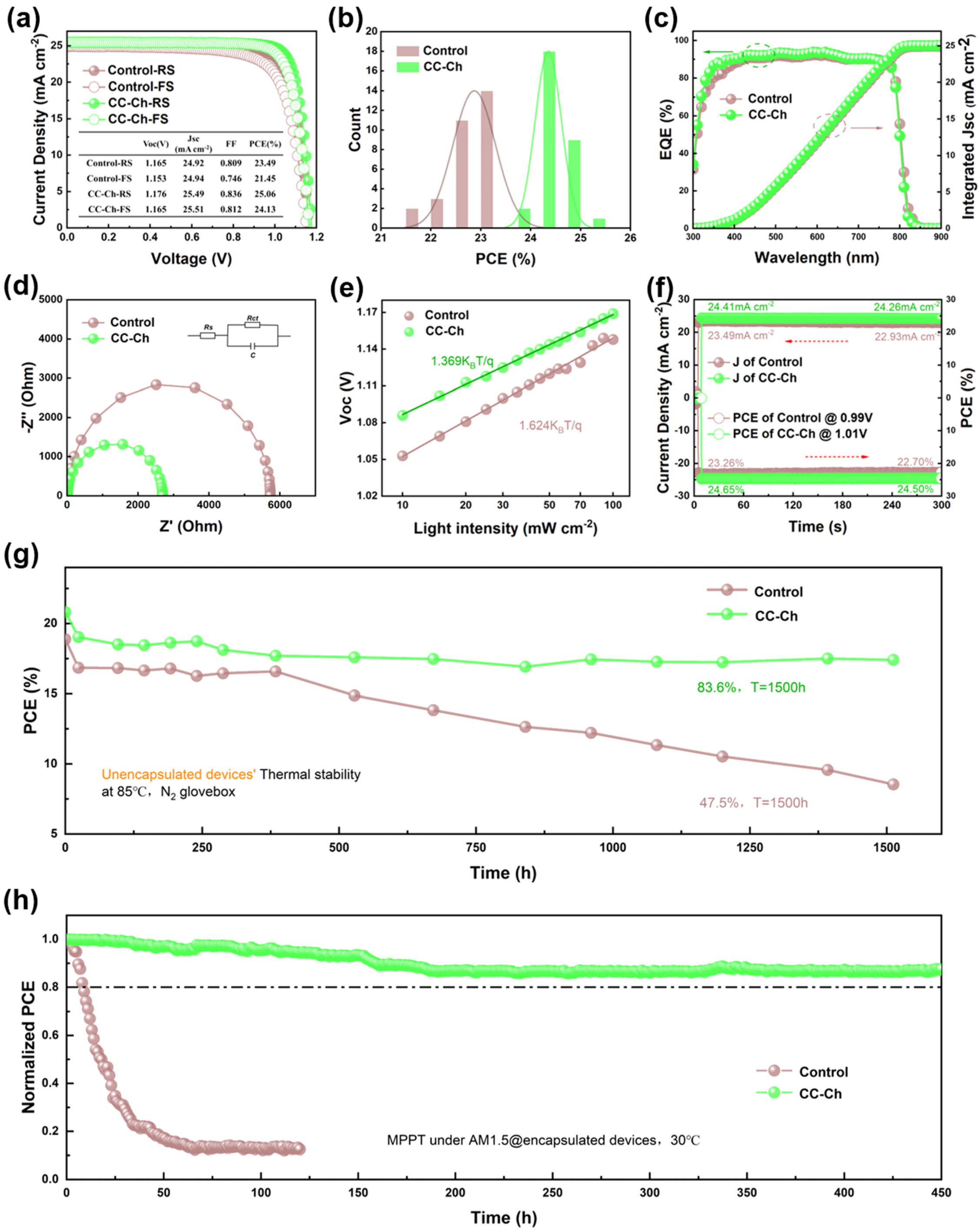

Fig. 6 (a) J–V curves of champion devices with or without Ch. (b) Statistics of PCE distribution of 30 devices with or without Ch addition. (c) EQE and integrated current density spectra, (d) Nyquist plots, (e) V_{OC} *versus* light intensity plots, and (f) steady-state power output curves for the PSCs with or without Ch. (g) Thermal stability of unencapsulated control and target devices. (h) MPP tracking of the encapsulated control and target devices using PTAA as the HTL under AM1.5G illumination (100 mW cm^{-2}) in ambient air with a relative humidity of 25 ± 5% at 30 °C.

decreases and stabilizes at 24.50% at a bias voltage of 1.01 V, while the output power of the control PSCs decreases substantially to 22.70% at a bias voltage of 0.99 V.

To evaluate the thermal stability of the PSCs, we measured the evolution of the PCE of the unencapsulated devices under a nitrogen atmosphere at 85 °C. As shown in Fig. 6(g), after 1500 h, the PSCs with Ch still maintained about 83.6% of the initial PCE, while the control PSCs only maintained about 47.5% of the initial PCE. To further evaluate the long-term operational stability of the devices, we performed MPP tracking of the encapsulated devices under continuous AM1.5G illumination in an air environment. As shown in Fig. 6(h), the control

PSCs rapidly decayed to 17.3% of the initial PCE within 50 hours. The PSCs with Ch maintained 87.6% of the initial PCE after 450 h of continuous MPP tracking, which implies that the enhanced crystal quality and the *in situ* passivation of Ch inhibit ion migration and significantly improve both the thermal and light soaking stability of the PSC.[48] In addition, we applied the CC–Ch strategy to both 1 cm^2 devices and 5 cm × 5 cm mini-modules (aperture area: 10 cm^2) with the primary PCE of 21.58% (Fig. S15, ESI†) and 20.08% (Fig. S16, ESI†), respectively. These results suggest that the reported catalyzed crystallization is an effective and promising up-scalable route for preparing high-quality α-$FAPbI_3$ perovskite and high-performance solar cells.

3. Conclusion

In summary, we have presented a novel strategy of catalysis crystallization to prepare α-$FAPbI_3$ perovskite films with the tricyclic alkaloid colchicine for antisolvent-free PSCs. The strong coordination interactions between Ch (CONH) and Pb^{2+}, hydrogen bonding between Ch (C=O) and FA^+, and hydrogen bonding forces between Ch (N–H) and I, assist the direct nucleation and crystallization of α-$FAPbI_3$ without intermediate phase transition of δ-$FAPbI_3$. Such strong interactions can also *in situ* passivate the uncoordinated Pb^{2+} and I^- vacancies, inhibit the precipitation of metal Pb^0, reduce the nonradiative recombination, increase the carrier lifetime, and improve the device stability. In addition, Ch can improve the energy level alignment of the device, which greatly optimizes the carrier extraction and transport. As a result, a champion PCE of 25.06% was achieved for the devices with the addition of Ch. Notably, the Ch-based unencapsulated PSCs maintained an initial efficiency of 83.6% after aging at 85 °C under a nitrogen atmosphere for 1500 hours. The MPP tracking of the encapsulated devices under continuous AM1.5G illumination in an air environment maintained 87.6% of the initial PCE after 450 h. This work provides an effective route of using natural green molecules to facilitate the crystallization of α-$FAPbI_3$ film in the antisolvent-free PSCs, further benefiting the upscaling and commercialization of high-performance perovskite solar cells.

Author contributions

J. Z. supervised the project. J. Z. and Z. Z. W. conceived the idea, designed the experiments, and interpreted the data. Z. Z. W. and H. Y. C. contributed equally to this work. Z. Z. W., H. Y. C., T. W., and J. Y. X. designed and optimized the experimental conditions. Z. Z. W., H. Y. C., Z. Y. W., and H. Q. D. carried out the experiments for the device analysis. J. Z., F. Z. H., Y. B. C., and J. Z. provided helpful discussions for paper writing. All the authors made substantial contributions to the discussion of the content and reviewed and edited the manuscript before submission. Z. Z. W., H. Y. C. and J. Z. wrote the manuscript with input from all the authors.

Conflicts of interest

The authors declare no competing interests.

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References

1 A. Kojima, K. Teshima, Y. Shirai and T. Miyasaka, *J. Am. Chem. Soc.*, 2009, **131**, 6050–6051.
2 NREL, Best Research-Cell Efficiency Chart, https://www.nrel.gov/pv/cell-efficiency.html, (accessed August, 2023).
3 S.-H. Turren-Cruz, A. Hagfeldt and M. Saliba, *Science*, 2018, **362**, 449–453.
4 B.-w Park, H. W. Kwon, Y. Lee, D. Y. Lee, M. G. Kim, G. Kim, K.-j Kim, Y. K. Kim, J. Im, T. J. Shin and S. I. Seok, *Nat. Energy*, 2021, **6**, 419–428.
5 P. Shi, Y. Ding, B. Ding, Q. Xing, T. Kodalle, C. M. Sutter-Fella, I. Yavuz, C. Yao, W. Fan, J. Xu, Y. Tian, D. Gu, K. Zhao, S. Tan, X. Zhang, L. Yao, P. J. Dyson, J. L. Slack, D. Yang, J. Xue, M. K. Nazeeruddin, Y. Yang and R. Wang, *Nature*, 2023, **620**, 323–327.
6 W. Hui, L. Chao, H. Lu, F. Xia, Q. Wei, Z. Su, T. Niu, L. Tao, B. Du, D. Li, Y. Wang, H. Dong, S. Zuo, B. Li, W. Shi, X. Ran, P. Li, H. Zhang, Z. Wu, C. Ran, L. Song, G. Xing, X. Gao, J. Zhang, Y. Xia, Y. Chen and W. Huang, *Science*, 2021, **371**, 1359–1364.
7 J. J. Yoo, G. Seo, M. R. Chua, T. G. Park, Y. Lu, F. Rotermund, Y.-K. Kim, C. S. Moon, N. J. Jeon, J.-P. Correa-Baena, V. Bulović, S. S. Shin, M. G. Bawendi and J. Seo, *Nature*, 2021, **590**, 587–593.
8 T. Bu, J. Li, H. Li, C. Tian, J. Su, G. Tong, L. K. Ono, C. Wang, Z. Lin, N. Chai, X.-L. Zhang, J. Chang, J. Lu, J. Zhong, W. Huang, Y. Qi, Y.-B. Cheng and F. Huang, *Science*, 2021, **372**, 1327–1332.
9 Y. Zhao, F. Ma, Z. Qu, S. Yu, T. Shen, H.-X. Deng, X. Chu, X. Peng, Y. Yuan, X. Zhang and J. You, *Science*, 2022, **377**, 531–534.
10 X. Liu, D. Luo, Z.-H. Lu, J. S. Yun, M. Saliba, S. I. Seok and W. Zhang, *Nat. Rev. Chem.*, 2023, **7**, 462–479.
11 J. Park, J. Kim, H.-S. Yun, M. J. Paik, E. Noh, H. J. Mun, M. G. Kim, T. J. Shin and S. I. Seok, *Nature*, 2023, **616**, 724–730.
12 S. Wang, T. Yang, Y. Yang, Y. Du, W. Huang, L. Cheng, H. Li, P. Wang, Y. Wang, Y. Zhang, C. Ma, P. Liu, G. Zhao, Z. Ding, S. Liu and K. Zhao, *Adv. Mater.*, 2023, **35**, 2305314.
13 Y. Meng, Y. Wang, C. Liu, P. Yan, K. Sun, Y. Wang, R. Tian, R. Cao, J. Zhu, H. Do, J. Lu and Z. Ge, *Adv. Mater.*, 2024, **36**, 2309208.

14 Y. Hu, Q. Zhong, B. Song, H. Xu, Q. Li, S. Li, Y. Qiu, X. Yang, J. Chen, Q. Zhang, R. Zhu and M. Cao, *Angew. Chem., Int. Ed.*, 2024, **63**, e202316154.
15 R. Wang, X. Li, J. Qi, C. Su, J. Yang, S. Yang, M. Yuan and T. He, *Adv. Mater.*, 2023, **35**, 2304149.
16 R. Tian, C. Liu, Y. Meng, Y. Wang, R. Cao, B. Tang, D. Walsh, H. Do, H. Wu, K. Wang, K. Sun, S. Yang, J. Zhu, X. Li and Z. Ge, *Adv. Mater.*, 2023, 2309998.
17 X. Du, J. Zhang, H. Su, X. Guo, Y. Hu, D. Liu, N. Yuan, J. Ding, L. Gao and S. Liu, *Adv. Mater.*, 2022, **34**, 2204098.
18 M. Li, R. Sun, J. Chang, J. Dong, Q. Tian, H. Wang, Z. Li, P. Yang, H. Shi, C. Yang, Z. Wu, R. Li, Y. Yang, A. Wang, S. Zhang, F. Wang, W. Huang and T. Qin, *Nat. Commun.*, 2023, **14**, 573.
19 A. D. Taylor, Q. Sun, K. P. Goetz, Q. An, T. Schramm, Y. Hofstetter, M. Litterst, F. Paulus and Y. Vaynzof, *Nat. Commun.*, 2021, **12**, 1878.
20 H.-H. Huang, Q.-H. Liu, H. Tsai, S. Shrestha, L.-Y. Su, P.-T. Chen, Y.-T. Chen, T.-A. Yang, H. Lu, C.-H. Chuang, K.-F. Lin, S.-P. Rwei, W. Nie and L. Wang, *Joule*, 2021, **5**, 958–974.
21 Y. Ren, J. Chen, D. Ji, Y. Sun and C. Li, *Chem. Eng. J.*, 2020, **384**, 123273.
22 W. Shen, H. Cai, Y. Kong, W. Dong, C. Bai, G. Liang, W. Li, J. Zhao, F. Huang, Y.-B. Cheng and J. Zhong, *Small*, 2023, **19**, 2302194.
23 J. W. Yoo, J. Jang, U. Kim, Y. Lee, S.-G. Ji, E. Noh, S. Hong, M. Choi and S. I. Seok, *Joule*, 2021, **5**, 2420–2436.
24 H.-S. Yun, H. W. Kwon, M. J. Paik, S. Hong, J. Kim, E. Noh, J. Park, Y. Lee and S. Seok II, *Nat. Energy*, 2022, **7**, 828–834.
25 G. Li, Z. Wang, Y. Wang, Z. Yang, P. Dong, Y. Feng, Y. Jiang, S.-P. Feng, G. Zhou, J.-M. Liu and J. Gao, *Small*, 2023, **19**, 2301323.
26 L. Chao, Y. Xia, X. Duan, Y. Wang, C. Ran, T. Niu, L. Gu, D. Li, J. Hu, X. Gao, J. Zhang and Y. Chen, *Joule*, 2022, **6**, 2203–2217.
27 X. Wu, Y. Zheng, J. Liang, C. Tian, A. Sun, C. Tang, Y. Liu, S. Zhang, C. Wang, S. Song, C.-C. Chen and K. M. Reddy, *Adv. Energy Mater.*, 2024, 2304302.
28 T. Niu, L. Chao, Y. Xia, K. Wang, X. Ran, X. Huang, C. Chen, J. Wang, D. Li, Z. Su, Z. Hu, X. Gao, J. Zhang and Y. Chen, *Adv. Mater.*, 2023, 2309171.
29 D. Wang, J. Chen, P. Zhu, Y. Qiao, H. Hu, J. Zeng, J. Zhang, G. Qu, Y. Wang, X. Wang, A. K.-Y. Jen and B. Xu, *Adv. Energy Mater.*, 2023, **13**, 2203649.
30 S. Yang, J. Dai, Z. Yu, Y. Shao, Y. Zhou, X. Xiao, X. C. Zeng and J. Huang, *J. Am. Chem. Soc.*, 2019, **141**, 5781–5787.
31 E. Radicchi, E. Mosconi, F. Elisei, F. Nunzi and F. De Angelis, *ACS Appl. Energy Mater.*, 2019, **2**, 3400–3409.
32 K. G. Stamplecoskie, J. S. Manser and P. V. Kamat, *Energy Environ. Sci.*, 2015, **8**, 208–215.
33 X. Li, W. Wang, K. Wei, J. Deng, P. Huang, P. Dong, X. Cai, L. Yang, W. Tang and J. Zhang, *Adv. Mater.*, 2023, 2308969.
34 B. Liu, X. Ren, R. Li, Y. Chen, D. He, Y. Li, Q. Zhou, D. Ma, X. Han, X. Shai, K. Yang, S. Lu, Z. Zhang, J. Feng, C. Chen, J. Yi and J. Chen, *Adv. Mater.*, 2024, 2312679.
35 C. Li, X. Wang, E. Bi, F. Jiang, S. M. Park, Y. Li, L. Chen, Z. Wang, L. Zeng, H. Chen, Y. Liu, C. R. Grice, A. Abudulimu, J. Chung, Y. Xian, T. Zhu, H. Lai, B. Chen, R. J. Ellingson, F. Fu, D. S. Ginger, Z. Song, E. H. Sargent and Y. Yan, *Science*, 2023, **379**, 690–694.
36 C. Liu, H. Su, Y. Pu, M. Guo, P. Zhai, Z. Liu and Z. Zhang, *Nano Energy*, 2023, **118**, 108990.
37 R. Wang, J. Xue, K.-L. Wang, Z.-K. Wang, Y. Luo, D. Fenning, G. Xu, S. Nuryyeva, T. Huang, Y. Zhao, J. L. Yang, J. Zhu, M. Wang, S. Tan, I. Yavuz, K. N. Houk and Y. Yang, *Science*, 2019, **366**, 1509–1513.
38 Z. Huang, Y. Bai, X. Huang, J. Li, Y. Wu, Y. Chen, K. Li, X. Niu, N. Li, G. Liu, Y. Zhang, H. Zai, Q. Chen, T. Lei, L. Wang and H. Zhou, *Nature*, 2023, **623**, 531–537.
39 O. J. Weber, D. Ghosh, S. Gaines, P. F. Henry, A. B. Walker, M. S. Islam and M. T. Weller, *Chem. Mater.*, 2018, **30**, 3768–3778.
40 B.-w Park and S. I. Seok, *Adv. Mater.*, 2019, **31**, 1805337.
41 Z.-A. Nan, L. Chen, Q. Liu, S.-H. Wang, Z.-X. Chen, S.-Y. Kang, J.-B. Ji, Y.-Y. Tan, Y. Hui, J.-W. Yan, Z.-X. Xie, W.-Z. Liang, B.-W. Mao and Z.-Q. Tian, *Chem*, 2021, **7**, 2513–2526.
42 Q. Cao, J. Yang, T. Wang, Y. Li, X. Pu, J. Zhao, Y. Zhang, H. Zhou, X. Li and X. Li, *Energy Environ. Sci.*, 2021, **14**, 5406–5415.
43 X. Yuan, R. Li, Z. Xiong, P. Li, G. O. Odunmbaku, K. Sun, Y. Deng and S. Chen, *Adv. Funct. Mater.*, 2023, **33**, 2215096.
44 J. S. Yun, A. Ho-Baillie, S. Huang, S. H. Woo, Y. Heo, J. Seidel, F. Huang, Y.-B. Cheng and M. A. Green, *J. Phys. Chem. Lett.*, 2015, **6**, 875–880.
45 H. Min, S.-G. Ji and S. I. Seok, *Joule*, 2022, **6**, 2175–2185.
46 Q. Han, S.-H. Bae, P. Sun, Y.-T. Hsieh, Y. Yang, Y. S. Rim, H. Zhao, Q. Chen, W. Shi, G. Li and Y. Yang, *Adv. Mater.*, 2016, **28**, 2253–2258.
47 N. Wu, T. Yang, Z. Wang, Y. Wu, Y. Wang, C. Ma, H. Li, Y. Du, D. Zhao, S. Wang, P. Liu, W. Huang, X. Ren, S. Liu and K. Zhao, *Adv. Mater.*, 2023, **35**, 2304809.
48 R. Wang, J. Xue, L. Meng, J.-W. Lee, Z. Zhao, P. Sun, L. Cai, T. Huang, Z. Wang, Z.-K. Wang, Y. Duan, J. L. Yang, S. Tan, Y. Yuan, Y. Huang and Y. Yang, *Joule*, 2019, **3**, 1464–1477.